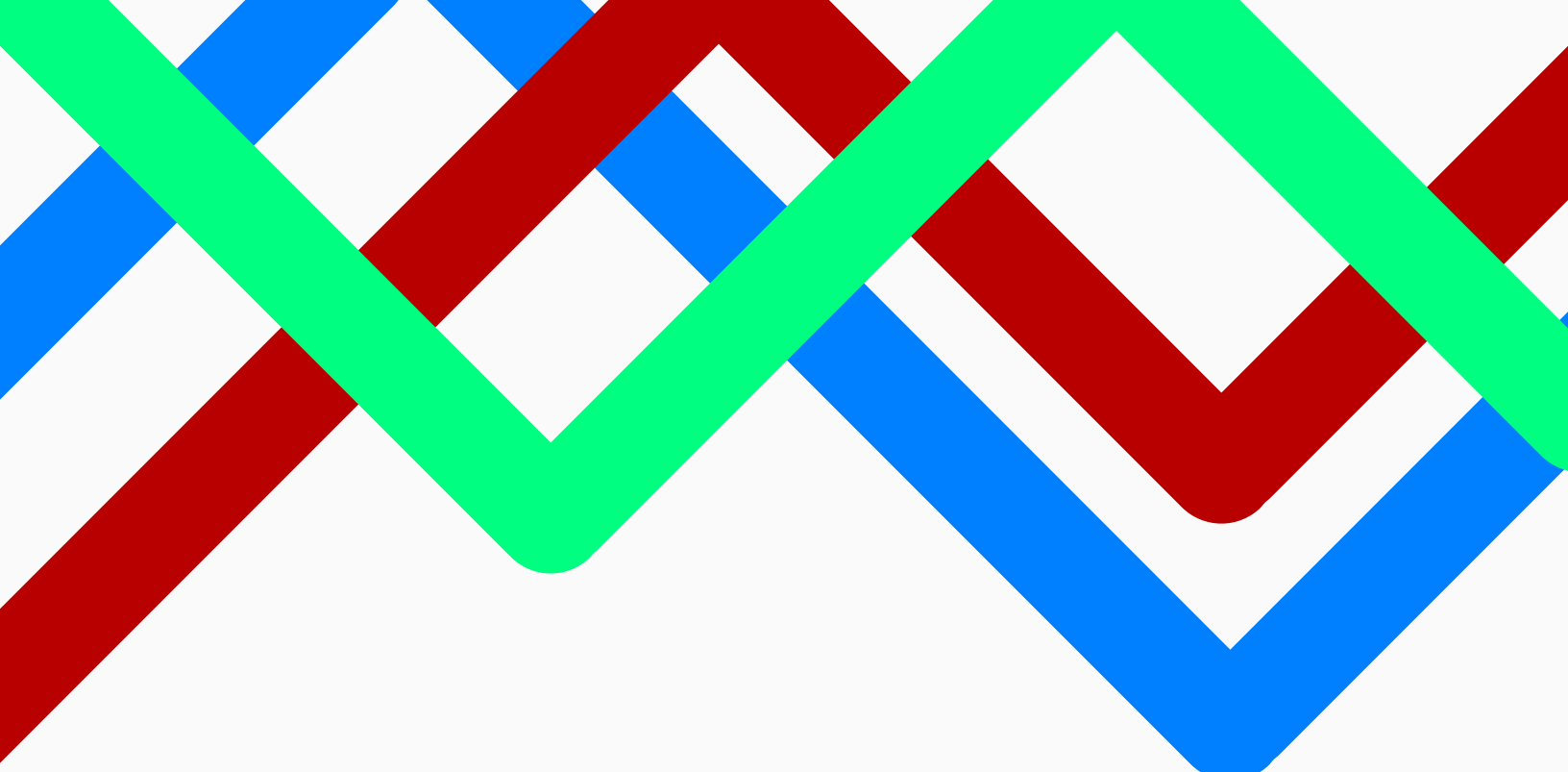




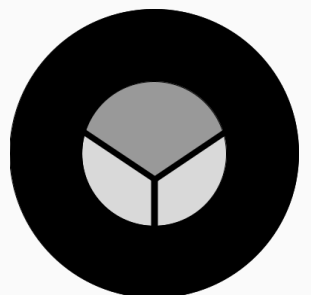
Cracker Protein Lab

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DE Intro to Biology I

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“Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things” - Issac Newton



1 Abstract

To show the denaturing of amylase through the change in pH levels and how the efficacy and efficiency of enzyme performance are scale-based.

2 Background / Introduction

3 Question / Problem

How does ***amylase***, an enzyme found in saliva, break polysaccharides into simpler carbohydrates in different PH levels?

4 Elements & Hypothesis

4.1 Hypothesis

If the pH levels of three solutions of pH {4, 5.5, 7}, are exposed to the enzyme *amylase* found in saliva and mixed with a polysaccharide containing cracker, then there should be a trend whereas the pH 7 solution will have the highest density of glucose, the pH 5.5 solution should have the medium density of glucose, and the pH 4 solution should have the lowest density of glucose.

4.2 Experimental Elements

- Independent Variable: PH Level of the solution
- Dependent Variable: Glucose concentration produced (in mmol/L)
- Experimental Groups:
 - Group 1: Saltine Cracker, Saliva, pH 4.0 solution
 - Group 2: Saltine Cracker, Saliva, pH 5.5 solution
- Positive Control Group:
 - Saltine Cracker, Saliva, optimal pH solution (pH 7.0)
- Negative Control Group:
 - Saltine Cracker, optimal pH solution (pH 7.0)
- Controls
 - Saliva to Solution ratio
 - Cracker to Solution ratio
 - Time allowed for enzymatic processes
 - Method of glucose measurement

5 Materials / Set up

5.1 Materials

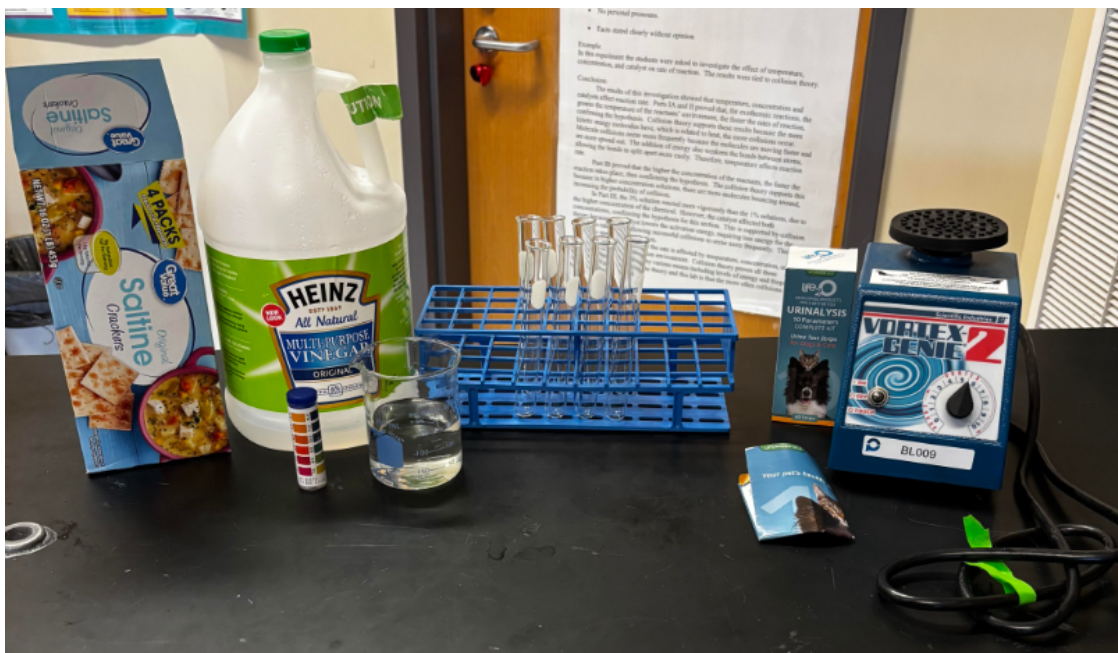
The following materials were gathered:

1. 4 saltine cracker pieces (approximately 1 gram each)
2. Roughly 100 mL of Heinz all-natural vinegar
3. pH test strips
4. Roughly 250 mL of distilled water
5. 8 15-25mL test tubes
6. Test tube rack
7. Life2O Pet Urinalysis Test Strips: Dog and Cat Urine Test Strips
8. Genie Vortex-Genie 2 Vortex Mixer SI-0236
9. Fresh Human Saliva (Roughly 20 mL total)
10. Stopwatch
11. Plastic pipettes
12. 4 stirring rods
13. Plastic gloves
14. Tape
15. Black Marker
16. 2x 250 mL beaker
17. Plastic test tubes
18. Mortar & Pestle

5.2 Set Up Procedure

1. All lab materials were collected and thoroughly inspected to ensure cleanliness.
2. Two sets of four test tubes were labeled with {"C","pH=7","pH=5.5","pH=4"} using tape and a marker.
3. The first set of four test tubes were filled with 10 mL of distilled water.
 - The water was filled up in the first beaker with about 100 mL of distilled water.
 - The vinegar was filled up to about the 50 mL mark of the second beaker.
4. Then, according to the test tube labels, the first set of four had 2 mL of vinegar added incrementally (until the pH was correct) and checked via a pH strip.
5. When all four test tubes of the first set were at the pH level indicated by each respective label, the tubes were put away, in a row, back onto the test tube rack

An exemplar image of an example setup without pH solutions can be found below:



6 Methods / Procedure

1. Saliva was collected by each individual labmate into their respective plastic test tube. This was to minimize the safety impact of having different contaminants on each test tube.
- 2.

7 Data / Results

Group	Glucose Measured (mmol/L)
Negative Control	0
Positive Control	15
pH=5.5	5
pH=4.0	0

8 Conclusion / Discussion